THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS UNIT TEST SET A

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The value of 52^2 is: 1

(a) 2756 **(b)** 2652 **(c)** 2714 **(d)** 2704

2. The area of a square of side $(x + 2)$ is: 1

(a) $4x + 4$ **(b)** $x^2 + 4$ **(c)** $x^2 + 2x + 4$ **(d)** $x^2 + 4x + 4$

3. If $a = 3$ and $b = 5$, then $(a + b)^2$ equals: 1

(a) 4 **(b)** 49 **(c)** 64 **(d)** 34

4. A car covers 240 km in 4 hours. In 7 hours at the same speed it covers: 1

(a) 480 km **(b)** 360 km **(c)** 420 km **(d)** 280 km

5. 3 hours in minutes is: 1

(a) 180 minutes **(b)** 30 minutes **(c)** 360 minutes **(d)** 300 minutes

6. A mixture has milk and water in the ratio 3 : 2. In 20 L of mixture, milk is: 1

(a) 12 L **(b)** 10 L **(c)** 8 L **(d)** 15 L

7. An article bought for ₹ 800 is sold for ₹ 720. The loss percent is: 1

(a) 5% **(b)** 10% **(c)** 80% **(d)** 15%

8. The marked price is ₹ 1500 and the discount is 20%. The selling price is: 1

(a) ₹ 1200 **(b)** ₹ 1800 **(c)** ₹ 1480 **(d)** ₹ 300

9. Multiplying a Pythagorean triplet by a constant gives: 1

(a) a prime triplet **(b)** a non-triplet
(c) a square **(d)** another Pythagorean triplet

10. The hypotenuse of a right triangle is 41 cm and one leg is 9 cm. The other leg is: 1

(a) 40 cm **(b)** 32 cm **(c)** 39 cm **(d)** 41 cm

11. The map distance is 5 cm and the actual distance is 20 km. The scale is 1 cm = 1

(a) 0.25 km **(b)** 100 km **(c)** 4 km **(d)** 5 km

12. If the speed doubles, the time taken for the same distance: 1

(a) halves **(b)** triples **(c)** doubles **(d)** remains same

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 10–15 words each.

13. Expand $(x + 6)^2$. 2

14. Write the ratio of 75 paise to ₹ 3 in simplest form. 2

15. Express 35 out of 140 as a percentage. 2

16. Is (10, 24, 26) a Pythagorean triplet? Verify. 2

17. Write the ratio 2 : 3 and 3 : 5 as a single three-term ratio. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 30–40 words each.

18. Find 304^2 using a suitable identity. 3

19. A trader buys an article for ₹ 2500 and sells at 6% profit. Find the selling price. 3

20. A rectangular park measures 7 m by 24 m. A boy walks along the diagonal instead of two sides. How much distance does he save? 3

21. On a map, 1 cm stands for 50 km. Two cities are 6.8 cm apart. Find the actual distance. How long does a train at 60 km/h take? 3

22. A ladder of length 29 m rests against a wall with its foot 20 m from the wall. How high up the wall does it reach? 3

OR

Using $(n,\ \frac{n^2-1}{2},\ \frac{n^2+1}{2})$ find the triplet for $n = 3$ and verify. 3

23. Divide ₹ 1800 among A, B and C in the ratio 2:3:4. 3

OR

A map scale shows 5 cm = 20 km. What is 1 cm in km, and what is the representative ratio? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

24. **Case study:** A school's 480 students are in the ratio 5 : 3 for Primary and Secondary sections. 5

(i) How many in each?

(ii) If 20 Primary students shift to Secondary, find the new ratio.

(iii) Comment.

OR

(i) Convert 36 km/h to m/s. 5

(ii) A cyclist travels 15 km in 50 minutes; find speed in km/h.

25. **Case study:** A pie chart shows the time spent by a student in a 24-hour day: sleep 8 h, school 6 h, study 4 h, play 3 h, other 3 h. 5

(i) Find each angle.

(ii) Which sector is the largest?

(iii) By how many degrees does study exceed play?

OR

Case study: A model of a building is made at a scale of 1 : 150. 5

(i) A 45 m tall building is how tall on the model?

(ii) The model's door is 1.6 cm; find the real door.

(iii) Floor area ratio between model and real.

END OF QUESTION PAPER — Total: 50 Marks